



Solution Guide: Service Users

In the previous activity, we stopped and removed a few old services from the system. In this activity, we removed those users from the system and added a new service user for tripwire.

To complete this activity, we needed to:

- Use the `deluser` command to remove lingering service users;
- Use the `adduser` command with the correct flags to create a new tripwire user;
- Edit the `sudoers` file to allow the tripwire user to run tripwire with `sudo`; and
- Change the tripwire permissions to only allow the owner of tripwire to run the service.

Note: These steps are not always needed, as most services create their own user when the package is installed.

Solution

Note: The bonus solution is included.

1. The first step is to remove any service users associated with the following services: `ftp` and `dovecot`:
 - We can quickly find these users with `grep "ftp\|dove" /etc/passwd`.
 - To remove the service users, run `sudo deluser --remove-all-files <username>` for each user.
 - For example: `sudo deluser --remove-all-files dovecot`
2. We will create a tripwire user that will be dedicated to running Tripwire:
 - Run `sudo adduser --system --no-create-home tripwire`.
 - Run `id tripwire` and verify that the UID is less than 1,000.
 - Run `ls /home` to verify there is no tripwire home folder.
3. Remember, we can observe password entries in the `/etc/shadow` file.
 - Run `sudo tail /etc/shadow`.
4. The `*` in the password field for the Tripwire user means the user is locked without a password.
 - Run `sudo tail /etc/passwd`.
5. Note that `usr/sbin/nologin` is at the end of the Tripwire line.
6. We will add a line to the `sudoers` file in order to allow this user to run only tripwire using `sudo` privileges.
 - Run `sudo visudo`.

- Add `tripwire ALL= NOPASSWD: /usr/sbin/tripwire` to the user section of the file and save it.
- The section should be as follows:

User privilege specification

root ALL=(ALL:ALL) ALL

- `tripwire ALL= NOPASSWD: /usr/sbin/tripwire`

7. We will change the permission of the `tripwire` program to only allow the owner to execute it.

- Run `which tripwire` to locate the `tripwire` package.
- Run `sudo chmod 700 /usr/sbin/tripwire`.
- Run `ls -l /usr/sbin/tripwire` to verify.